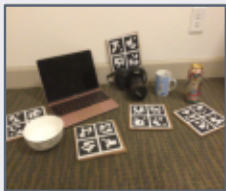


Input



RGB image



instance mask

Geometry model (MoT)

shape branch

* frozen

layout branch

▲ trained



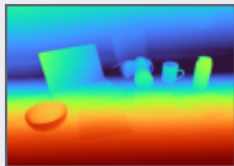
canonical shape, extents $\mathbf{c}$

SSI ratio $\tilde{\mathbf{s}}$, translation
(dimensionless)

MetricScaleHead

▲ trained

Metric anchor



MoGe-2

* frozen

metric pointmap $\mathbf{P}$
 S_{scene} : units live here

swappable: MoGe-v1 (affine) / MoGe-2 / sensor

SLAT refinement

scale token

cross-attention

▲ trained

self-attn + FFN

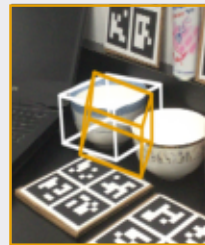
* frozen

scale token injected
into conditioning

MetricScaleDecoder

▲ trained

Metric output



metric 3D box (vs GT)

$\mathbf{d} = [W, H, D]$ metres

mesh $\times$ metric scale
+ pose (R, t)

$\mathbf{d} = \mathbf{c} \odot \tilde{\mathbf{s}} \cdot S_{\text{scene}}$

S_{scene} lifts SSI $\rightarrow$ metric (delegation)

pose (R, t)